Samuel King

PhD Student · Georgetown University

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Professional Summary

Fifth-year PhD candidate in theoretical computer science with research experience in algorithms and data structures. Data scientist with professional experience building machine learning tools in applied settings. Strong foundations in algorithm design, mathematical modeling, and deep learning. Interested in developing novel algorithmic solutions to complex data challenges.

Education

Georgetown University — *Washington D.C.*

PhD, Computer Science — *Fall 2021 - present*

- Advisor: Sasha Golovnev
- Research Interests: Algorithms, Data Structures, Discrete Math, Complexity
- GPA: 4.0

University of Rochester — *Rochester, NY*

Honors BS, Mathematics — *Fall 2016 - Spring 2021*

- Computer Science Minor
- Thesis: Sampling Young Tableaux and Contingency Tables
- Take Five Scholar: The Role of Music in Video Games and Film
- Magna Cum Laude, GPA: 3.96

Work Experience

Predict Health, Inc. — *Arlington, VA*

Data Scientist — *May 2025 - Present*

- Researched and developed retention models for Medicare populations to expand Predict Health's data science capabilities
- Built a custom Python package to ensure efficient, reusable capabilities for future retention projects
- Streamlined codebase to reduce project completion time from months to days
- Designed experiments to detect and mitigate data leakage in temporal prediction tasks
- Developed efficient pipeline for matching 100M records in Predict Health's data lake

Georgetown University — *Washington, D.C.*

Course Designer and Instructor: *Intro to Programming in the Age of AI* — *January 2026 - Present*

- Designed curriculum and materials for Georgetown's first AI-integrated Intro to Programming course
- Led twice-weekly lectures and workshops to teach students AI coding assistant usage and python fundamentals

Publications & Patents

Karthik Gajulapalli, Alexander Golovnev, Samuel King, and Sidhant Saraogi. Online Orthogonal Vectors Revisited. *Symposium on Discrete Algorithms (SODA)*, 2026.

Karthik Gajulapalli, Alexander Golovnev, and Samuel King. On the Power of Adaptivity for Function Inversion. *Conference on Information-Theoretic Cryptography (ITC)*, 2024.

Aniruddha Bapat, Andrew M. Childs, Alexey V. Gorshkov, Samuel King, Eddie Schoute, and Hrishee Shastri. Quantum routing with fast reversals. *Quantum*, 5, 533, 2021.

U.S. Patent Application 18178491, filed Mar 3, 2023.
U.S. Patent Application 18669111, filed May 20, 2023.

Erin Bevilacqua, Samuel King, Jürgen Kritschgau, Michael Tait, Suzannah Tebon, and Michael Young. Rainbow numbers for $x_1 + x_2 = kx_3$ in $\mathbb{Z}_n$. *Integers: Electronic Journal of Combinatorial Number Theory*, 20(A50):A50, 2020.

Selected Coursework & Skills

Mathematics: Combinatorial Optimization, Spectral Graph Theory, Linear Algebra, Computational Statistics, Combinatorics, Algebra I & II
Computer Science: Algorithms, Data Structures, Sampling Algorithms, Analytical Methods, Cryptography, Parallel Algorithms, Databases, Gems of Theoretical CS
Artificial Intelligence: Experimental AI, Neural Networks, Artificial Intelligence, Philosophy of AI, AI Safety Fundamentals
Skills: Python (primary language), R, Mathematica, SQL (familiar languages), Git, LaTeX, PyTorch, NetworkX, Pandas, VSCode, Cursor, Databricks, Technical Communication

Projects

Reinforcement Learning for Ramsey Numbers *Georgetown University*
PERSONAL RESEARCH PROJECT *Summer 2024*

- Conducted research on RL approaches for finding counterexample graphs for lower bounds on Ramsey numbers
- Experimented with deep Q-learning, cross-entropy method, game-playing, and Monte Carlo tree search
- Experiments conducted using PyTorch, Gymnasium, Stable-Baselines3

Exploring Subitizing in Convolutional Neural Networks *Georgetown University*
NEURAL NETWORKS CLASS PROJECT *Fall 2022*

- Trained convolutional neural networks to count the number of simple shapes in an image
- Compared performance between classification and regression models
- Tested model robustness with different synthetic datasets

Stratego AI *University of Rochester*
HONORS SEMINAR RESEARCH PROJECT *Spring 2018*

- Implemented traditional state-space tree search algorithms to play the board game Stratego
- Experminted with using neural networks in Tensorflow to learn the state value function
- Experiments were unsuccessful, but the problem was solved by a team of 34 DeepMind scientists 4.5 years later

Teaching Experience

Intro to Programming in the Age of AI, Course Designer & Instructor *Georgetown U, Spring '26*
Gems of Theoretical CS, Teaching Assistant & Lecturer *Georgetown U, Fall '25*
CS Department Tutoring, Head Tutor & Program Founder *Georgetown U, Spring & Fall '25*
Data Structures, Guest Lecturer *Amherst College, Fall '24*
Introduction to Algorithms, Teaching Assistant & Lecturer *Georgetown U, Fall '24*
Math Methods for CS, Teaching Assistant & Lecturer *Georgetown U, Spring '24*
Intro to Databases, Teaching Assistant & Lecturer *Georgetown U, Fall '23*
New Horizons in Theoretical Computer Science, Teaching Assistant *TTIC (Virtual), Summer '23*
Graduate Algebra I, Teaching Assistant & Lecturer *U of Rochester, Fall '20*
Honors Calculus I & II, Teaching Assistant & Workshop Leader *U of Rochester, Fall '17-Spring '20*

Honors and Awards

2025 **Outstanding Teaching Assistant Award**, Georgetown U. Computer Science Department
2024 **Outstanding Teaching Assistant Award**, Georgetown U. Computer Science Department
2020 **Arthur S. Gale Memorial Prize**, University of Rochester Mathematics Department
2019 **Take Five Scholarship**, University of Rochester
Deans' Award for Symposium Presentation, University of Rochester
Joint Mathematics Meetings 2019 Outstanding Poster Distinction, JMM 2019